A microcontroller-driven class-D amplifier for a 1.7 kW subwoofer: prototype, design analysis and characterisation plan

Minka Dubourg

June 2024

Abstract

This report describes a full-bridge class-D amplifier designed to drive a 1.7 kW passive subwoofer for a student sound-system association that had no matching amplifier. An STM32L475 microcontroller samples the line-level input and generates 200 kHz pulse-width modulation. The PWM drives four 650 V super-junction MOSFETs on a custom milled H-bridge board.

A bench prototype validated the signal chain and reproduced audio through a small loudspeaker at reduced rail voltage. Its parts were the level shifter, the microcontroller PWM, the pre-amplifier and the H-bridge.

A datasheet-based design analysis then quantifies what the full-power design requires. On a 100 V rail, 1.7 kW needs a load of at most 2.94 Ω. The modelled efficiency at rated power is 90.6 %, but conduction losses would put 44 W in each device, beyond what the TO-220 FullPAK package can dissipate. At $\frac{1}{8}$ rated power, the selected 7 °C/W heatsinks are marginal.

The analysis also identifies three design issues and quantifies each. The output filter values as annotated would attenuate the subwoofer band by more than 20 dB. An op-amp gate drive cannot switch these MOSFETs at 200 kHz. Open-loop dead-time distortion is large at this switching frequency, about 10 % THD at 150 ns.

The firmware is redesigned to address the timing issues: four timer channels with hardware-placed dead time, ADC sampling synchronised to the PWM, and noise shaping that lifts the in-band SNR of the 7.6-bit modulator from 66 dB to 98 dB, where the 12 bit ADC becomes the limit. A twelve-point characterisation protocol defines how the targets are to be verified.

1 Introduction

The subwoofer to be driven is rated at 1.7 kW. It was built in-house, so no off-the-shelf amplifier came matched to it. The requirements were high efficiency (above 90 %), low parts cost, and room for later signal-processing features such as filtering, crossover and delay. Class D meets the efficiency requirement by operating the output transistors as switches rather than linear devices. A microcontroller-based modulator leaves room for the signal-processing features.

This report covers four things:

1. the architecture and a bench prototype of the full signal chain (Section 2);
2. a quantitative design analysis of the power stage: deliverable power, losses and thermal limits, gate drive, output filter, and dead-time distortion (Section 3);
3. a firmware redesign that addresses the timing requirements found in that analysis (Section 4);
4. a characterisation protocol for the full-power build (Section 5).

Results are labelled throughout as *measured* (prototype), *modelled* (datasheet-based calculation) or *simulated*.

2 Architecture and prototype

Signal chain. The line-level input (±1.23 V from a jack) is shifted into the 0 V to 3 V range of the microcontroller's ADC (Fig. 1). The STM32L475 runs at 80 MHz and compares each sample with a 200 kHz carrier to set the duty cycle. The bridge uses two-level (AD) modulation: leg A has duty d and leg B has duty $1 - d$, so the differential output is zero at $d = 0.5$. Each leg needs a high-side and a low-side gate signal, separated by a dead time so that the two switches of a leg never conduct together. The power stage is a full H-bridge of four NTPF082N65S3F MOSFETs on a 100 V rail, followed by a second-order LC filter.



Figure 1: Signal chain.

Block	Part	Key parameters
Modulator	STM32L475VG (B-L475E-IOT01A)	Cortex-M4F, 80 MHz, 12-bit ADC
Pre-amplifier (prototype)	TL081 op-amp	3.3 kΩ / 1 kΩ gain network
Switches	NTPF082N65S3F (×4)	650 V, $R_{DS(on)} \leq 82\,\text{m}\Omega$, $Q_g = 70\,\text{nC}$, $Q_{rr} = 397\,\text{nC}$
Heatsink	CO2245V1 (one per device)	$R_{\theta SA} = 7\,^\circ\text{C/W}$, TO-220
Rail	DC supply	100 V

Table 1: Main components.

Prototype. The level shifter, pre-amplifier and gate-drive stages were built on breadboard (Fig. 2). The H-bridge was built on a custom milled single-sided PCB with the four TO-220 devices and banana-jack power and output terminals. The full chain was assembled on the bench and reproduced audio through a small bookshelf loudspeaker at reduced rail voltage. In that test the loudspeaker's own inductance acted as the low-pass filter, because the LC filter was not populated.

Measured signals. Fig. 3 shows three oscilloscope captures:

- **(a) Level shifter output:** the shifted audio sits inside the ADC range (1.73 V peak-to-peak with an 82 Hz test tone).
- **(b) Microcontroller PWM output:** a 3.3 V-logic square wave at 195.8 kHz, which is 2 % below the 200 kHz nominal. The ringing on the edges is typical of a long probe ground lead.
- **(c) Two antiphase signals at** 197.8 kHz**, 7.0 V peak-to-peak:** their amplitude is above the 3.3 V logic level and their shape is nearly sinusoidal. That identifies them as the outputs of the op-amp pre-amplifier, which is limited in bandwidth and slew rate. The node was inferred from the waveform and was not recorded at capture time. Section 3.3 explains why this stage cannot drive the MOSFETs at full rail.

3 Design analysis

The prototype ran at low voltage. This section uses datasheet values [1] to determine what the 100 V, 1.7 kW design requires. All numbers come from `analysis/power_stage.py` and `analysis/pwm_sim.py`.

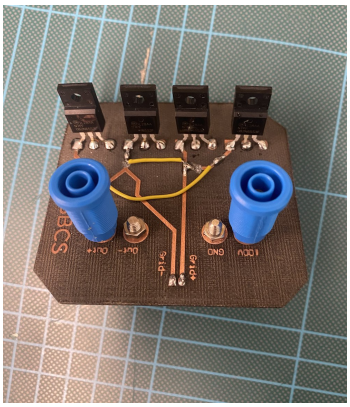
3.1 Deliverable power

A full bridge at full modulation delivers an undistorted sine power of $P = V_{\text{bus}}^2/2R$. On a 100 V rail that is 625 W into 8 Ω, 1250 W into 4 Ω and 2500 W into 2 Ω. The 1.7 kW target is therefore reachable only if the subwoofer's minimum impedance in band is at most 2.94 Ω. For an 8 Ω driver, the rail would have to rise to 165 V. Measuring $|Z(f)|$ is the first step of the protocol (test 0).

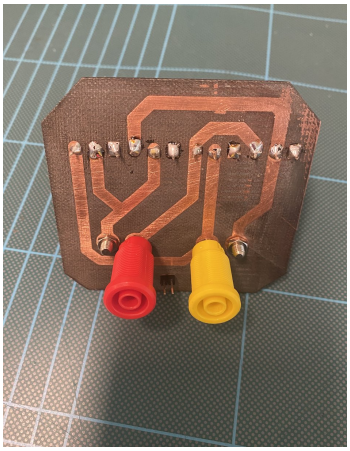
3.2 Losses, efficiency and thermal budget

The model accounts for:

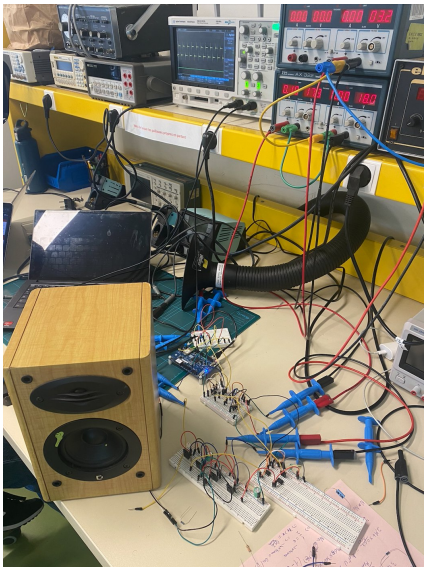
- conduction in the two devices carrying the load current, using $R_{DS(on)} = 82\,\text{m}\Omega \times 1.5$ at $T_J = 100\,^\circ\text{C}$ (datasheet Fig. 8);
- V–I overlap at each hard transition;
- body-diode reverse recovery;



(a) H-bridge board.



(b) Milled copper side.



(c) Bench audio test.

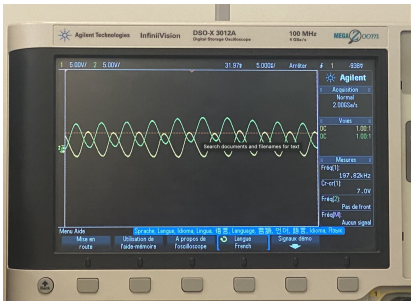
Figure 2: Prototype hardware.



(a) Level-shifter output.



(b) MCU PWM, 195.8 kHz.



(c) Pre-amplifier outputs.

Figure 3: Measured signals (oscilloscope captures).

- C_{oss} discharge ($E_{oss} = 3.5\,\mu\text{J}$ at 100 V);
- diode conduction during the dead time.

The recovery charge is scaled with the commutated current; a constant 397 nC gives a conservative bound.

Operating point (2.94 Ω)	Conduction	Overlap	Recovery	Other	Total	Per device
$\frac{1}{8}$ power, 212 W	17.8	4.6	6.0 (15.6)	2.6	30.9 (40.5)	7.7 (10.1)
Full power, 1.7 kW	142.2	13.0	16.8	4.7	176.8	44.2

Table 2: Modelled stage losses in W. Values in parentheses use the constant-Q_{rr} bound.

The modelled efficiency at rated power is 90.6 %, which meets the target. Conduction dominates (Table 2). Thermally, each device may dissipate at most

$$P_{\max} = \frac{T_{J,\max} - T_{\text{amb}}}{R_{\theta JC} + R_{\theta CS} + R_{\theta SA}} = \frac{150 - 45}{2.62 + 0.5 + 7} = 10.4\,\text{W}.$$

At $\frac{1}{8}$ power, a common test condition for music signals, the model gives 7.7 W to 10.1 W per device with $R_{DS(on)}$ taken at 100 °C, so the selected heatsinks are marginal. Near 150 °C, where the junction would actually sit, $R_{DS(on)}$ is higher still: with a factor of 1.9 to 2.2 the upper bound reaches 11.3 W to 12.2 W, above the limit. The exact factor should be read from the datasheet before the heatsinks are chosen. At continuous full power, the 44 W per device would raise the junction 116 °C above the case through $R_{\theta JC}$ alone, which no heatsink can accommodate.

Sustained full power would therefore need lower conduction loss. A MOSFET rated for the actual 100 V rail, rather than 650 V, typically offers several times lower $R_{DS(on)}$ and recovery charge. Conduction loss scales linearly with $R_{DS(on)}$.

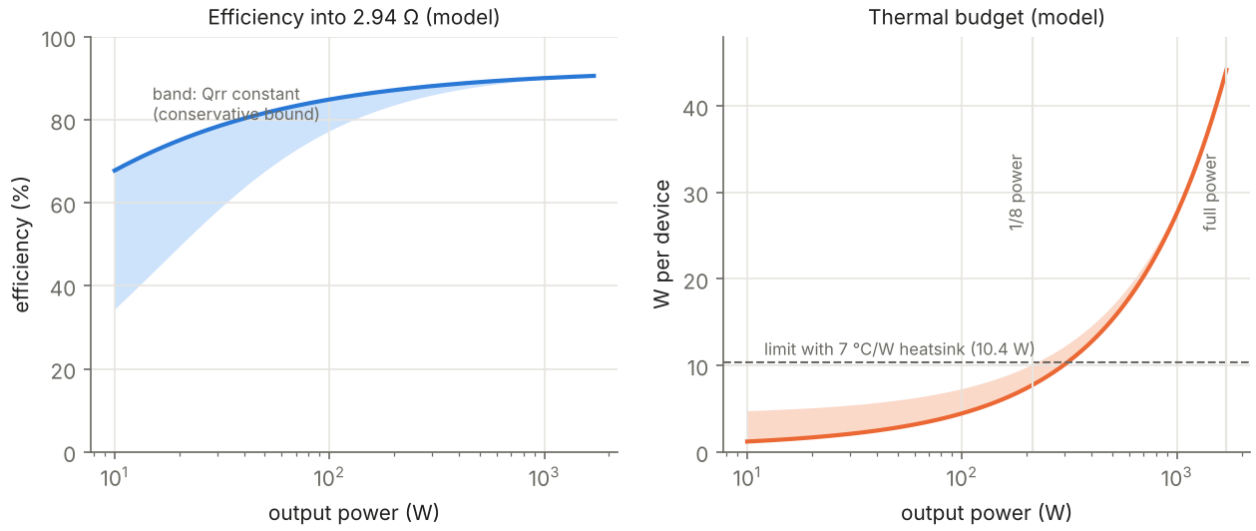


Figure 4: Modelled efficiency (left) and per-device dissipation (right) against output power into 2.94 Ω. The shaded band spans the two Q_{rr} assumptions. Dashed line: the largest dissipation the 7 °C/W heatsink allows at 45 °C ambient.

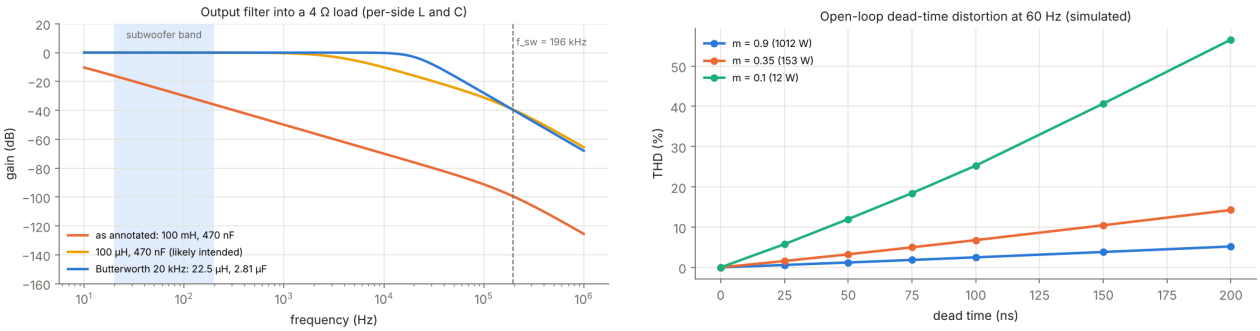


Figure 5: Left: transfer function of the output filter into 4 Ω. Right: simulated open-loop THD at 60 Hz against dead time, for three modulation depths.

3.3 Gate drive

Each MOSFET needs $Q_g = 70\,\text{nC}$ to switch. A general-purpose op-amp sources a few tens of milliamps, so charging the gate takes

$$t \approx \frac{Q_g}{I} \approx \frac{70\,\text{nC}}{20\,\text{mA}} = 3.5\,\mu\text{s},$$

which is longer than the 2.5 µs half-period at 200 kHz. The rounded 7 V waveform in Fig. 3c is consistent with this limit.

In addition, the two high-side N-channel devices need a gate voltage about 10 V above their source, which sits at the rail when they conduct. A ground-referenced driver cannot provide that once the rail exceeds a few volts. A dedicated half-bridge gate driver with a bootstrap supply and an ampere-level peak current would switch the gate in tens of nanoseconds. It is the first change required before raising the rail.

3.4 Output filter

The schematic annotates each side of the filter with 100 mH and 470 nF. Into a 4 Ω load, these values give a corner at 734 Hz, but the series inductance attenuates the subwoofer band itself: −24 dB at 50 Hz (Fig. 5). Such an inductor would also be impractically large at 24 A RMS. The values were most likely intended as 100 µH, which gives a well-behaved low-pass filter.

A second-order Butterworth design for 4 Ω at 20 kHz uses 22.5 µH and 2.81 µF per side. It is flat over 20 Hz to 200 Hz and attenuates the carrier by 40 dB.

3.5 Dead-time distortion and modulator resolution

During each dead time, the leg voltage is set by the direction of the load current rather than by the modulator. Averaged over a switching period, this adds an error voltage of $2V_{\mathrm{bus}} f_{\mathrm{sw}} t_{\mathrm{dead}} \operatorname{sign}(i)$ across the load. At 196 kHz and 150 ns, that is 5.9 % of the rail.

A cycle-averaged simulation through the Butterworth filter gives about 10 % THD at 153 W and much more at low level (Fig. 5). The dead time therefore has to be as short as the MOSFET turn-off allows ($t_{d(off)} + t_f \approx 70$ ns plus driver mismatch). Closed-loop feedback, which also improves PSRR, is the standard remedy. The original design plan already listed feedback as an improvement.

The PWM resolution is set by the timer: $f_{\mathrm{clk}}/f_{\mathrm{sw}} = 400$ counts per period, or about 7.6 bits of duty-cycle resolution in centre-aligned mode. Section 4 recovers in-band resolution with noise shaping.

4 Firmware redesign

The redesigned firmware (`firmware/`) is written against the CMSIS register definitions. It has four main features:

Gate timing. TIM2 runs in centre-aligned mode at 200 kHz. Its four channels drive the four gates, using Arduino-header pins D1, D0, D10 and D4. High-side channels use PWM mode 1 and low-side channels use PWM mode 2. Their compare values sit ±6 counts around the leg's switching point, which produces a 150 ns dead time on every edge without an advanced timer. The switching points are clamped so that every switch keeps a minimum on-time, which keeps a bootstrap supply charged.

Synchronised sampling. The timer's update event triggers the ADC twice per PWM period (400 kS/s). The conversion interrupt computes the next duty cycle, and preloaded compare registers apply it at the next update without glitches.

Signal processing. A one-pole DC tracker (about 1 Hz) removes the level-shifter offset without calibration. First-order error-feedback noise shaping moves the quantisation error of the 7.6-bit compare value above the audio band. On the host, the in-band (20 Hz to 200 Hz) SNR of the duty-cycle sequence for a 60 Hz tone rises from 65.5 dB with plain rounding to 98.2 dB with noise shaping. With an unquantised input it would reach 127 dB: once the PWM is noise-shaped, the 12 bit ADC sets the limit, and the real converter, whose effective resolution is below 12 bits, will set it lower still.

Start-up. The gate pins are held low as GPIO outputs. They are handed to the timer only after a 500 ms settling time, starting at 50 % duty so that the load sees zero volts.

The firmware builds with `arm-none-eabi-gcc` to 1.5 kB. Host-side unit tests check the timing invariants on 10^6 random inputs and the gain of a test tone; the dead time is checked on every sample. It has not yet been validated on hardware.

5 Characterisation plan

The targets are verified with the protocol in `docs/test-protocol.md`. It begins with a ramped power-up (12 V, 24 V, 48 V and 100 V) on a non-inductive dummy load and takes measurements with a differential probe. The protocol has twelve tests:

0. load impedance $|Z(f)|$;
1. switching signals, dead time and absence of shoot-through;
2. maximum power at 1 % THD+N;
3. efficiency $\eta(P_{\mathrm{out}})$;
4. THD+N against power and frequency;
5. frequency response;
6. residual noise and dynamic range;
7. switching residual at the load;

8. PSRR;
9. DC offset and turn-on transient;
10. thermal steady state at $\frac{1}{8}$ power;
11. a one-hour system test on the subwoofer.

Tests 3 and 4 are designed to calibrate the loss model and the dead-time simulation of Section 3 against measurements. A helper script computes THD+N and output power from oscilloscope or sound-card captures.

6 Conclusion and next steps

The prototype demonstrated the complete chain, from audio input through microcontroller PWM to the H-bridge and audible output, at reduced voltage. The design analysis shows that the 90 % efficiency target is plausible at rated power. It also shows that full power requires a load of at most 2.94 Ω and lower-$R_{DS(on)}$ devices for sustained operation.

The next steps, in order:

1. a bootstrap half-bridge gate driver;
2. protection before running at full rail voltage: a stalled or halted firmware freezes the duty cycle and can put DC across the subwoofer, so the design needs DC-offset detection with an output relay, a watchdog, and overcurrent shutdown (TIM1 and TIM8 have a hardware break input; TIM2 does not);
3. MOSFETs rated for the 100 V rail rather than 650 V;
4. a correctly dimensioned LC filter;
5. a two-layer PCB with a tight switching loop to reduce the ringing seen in Fig. 3b;
6. closed-loop feedback to suppress dead-time distortion and supply ripple;
7. the characterisation protocol of Section 5.

References

[1] onsemi. *NTPF082N65S3F Power MOSFET, N-Channel, SUPERFET III FRFET, 650 V, 40 A, 82 m*Ω. Datasheet rev. 4, 2018.

[2] Infineon Technologies. *Class D Audio Amplifier Design*. Training material, v01_00.

[3] Audio Engineering Society. *AES17-2020: AES standard method for digital audio engineering – Measurement of digital audio equipment.*